# Yizhou Zhou

yizhou82@student.ubc.ca | linkedin.com/in/yizhou-zhou | github.com/yizhou82

## Skills

**Programming Languages**: C/C++,C#, Python, Java, Assembly
**Development Tools**: Git/GitHub, Linux, SSH, Makefiles, GDB, ESP32, RTOS, ARM Cortex, NumPY/SciPy
**Communication Protocols**: SPI, I2C, UART, CAN, TCP/UDP
**Software & Applications**: Wireshark, VSCode, Altium, Arduino, PlatformIO, JIRA, Confluence, Microsoft Excel/Office

## Education

**University of British Columbia** — **Sept. 2021 – May 2026**
*Bachelor of Applied Science in Computer Engineering* — *Vancouver, BC*

- Courses: Real-time System Design, Operating Systems, Computer Architecture, Computer Networking, Cybersecurity

## Technical Experience

**Arlo Technologies** — **January 2025 – August 2025**
*Firmware Engineer Co-op* — *Richmond, BC*

- Integrated firmware solutions in C to improve security camera performance by understanding end-to-end system architecture and gaining exposure to embedded software development life cycle
- Developed privacy shutter and alert manager features in C by analyzing existing documentation and tracing system calls, along with working on bootloader code
- Debugged using system logs analysis and GDB for core dumps in Linux to resolve issues in a multi-submodule codebase
- Conducted 50+ Git deployments across 6 submodules, cherry-picking changes between branches, and resolving merge conflicts, while following sanity testing procedures
- Collaborated with 10 other engineers using JIRA and Confluence to manage development tickets and document technical workflows, improving team communication

**Sarcomere Dynamics** — **May 2024 – December 2024**
*Embedded Software Engineer Co-op* — *Vancouver, BC*

- Engineered solutions for robotic arms and hands in Python, developing low level drivers in C, connecting devices using SDKs and APIs, and processing data for teleoperative control
- Optimized state machine code in C with FreeRTOS for a PLC adapter to translate between digital and CAN signals
- Programmed STM32 microcontrollers in C++, configuring GPIO pinouts and interrupts; built and soldered circuits by following hardware specs and PCB schematics, testing with multimeters
- Tested CAN communication between ESP32 devices through the Arduino IDE in C and C++, using CppuTest for Unit Testing and logic analyzers for debugging CAN signals
- Developed network communication using Python and debugged with Wireshark, enabling reliable data transmission through radio link using UDP sockets over 150m in range

## Capstone Project

**Water Monitoring Device** — **September 2025 – April 2026**
*Project Manager/Firmware Developer* — *Vancouver, BC*

- Led a team of 5 in the development of a low cost IoT sensor to monitor water quality for underserved communities, focusing on developing firmware and mesh-net communication code
- Implemented system architecture and developed code on ESP32 for sensor integration and OTA capabilities, connecting devices using I2C and UART, while using FreeRTOS to manage scheduling and timing constraints
- Gained hands-on experience in TinyML and implementing wireless communication between ESP32s using ESP-NOW, BLE, and Lora, and creating MQTT communication with gateway node
- Designed PCB Board for gateway and central nodes using Altium, acquiring skills in schematic capture and routing

## Student Design Team

**UBC AgroBot** — **September 2023 – September 2024**
*Embedded Systems Developer* — *Vancouver, BC*

- Designed autonomous navigation software for the AgroBot, integrating real-time computer vision and motion planning algorithms in C++ to enable precise crop-row detection and path adjustment from live video streams
- Connected LiDAR data to a web server following websockets and networking protocols, using OpenCV to map the data
- Developed C++ modules and Makefiles for hardware integration in Linux to synchronize between hardware devices